

Wir verbinden den Buzzer direkt mit dem Arduino.  
GND zu Minus, Pin8 mit Plus.

Wir spielen ein Lied (Shape of you) ab mit folgendem Code:

```
// -----  
// Copyright (c) 2022 HiBit https://www.hibit.dev  
// -----  
  
#define BUZZER_PIN 9  
  
#define NOTE_B0 31  
  
#define NOTE_C1 33  
  
#define NOTE_CS1 35  
  
#define NOTE_D1 37  
  
#define NOTE_DS1 39  
  
#define NOTE_E1 41  
  
#define NOTE_F1 44  
  
#define NOTE_FS1 46  
  
#define NOTE_G1 49  
  
#define NOTE_GS1 52  
  
#define NOTE_A1 55  
  
#define NOTE_AS1 58  
  
#define NOTE_B1 62  
  
#define NOTE_C2 65  
  
#define NOTE_CS2 69  
  
#define NOTE_D2 73  
  
#define NOTE_DS2 78  
  
#define NOTE_E2 82  
  
#define NOTE_F2 87  
  
#define NOTE_FS2 93  
  
#define NOTE_G2 98  
  
#define NOTE_GS2 104  
  
#define NOTE_A2 110
```

```
#define NOTE_AS2 117
#define NOTE_B2 123
#define NOTE_C3 131
#define NOTE_CS3 139
#define NOTE_D3 147
#define NOTE_DS3 156
#define NOTE_E3 165
#define NOTE_F3 175
#define NOTE_FS3 185
#define NOTE_G3 196
#define NOTE_GS3 208
#define NOTE_A3 220
#define NOTE_AS3 233
#define NOTE_B3 247
#define NOTE_C4 262
#define NOTE_CS4 277
#define NOTE_D4 294
#define NOTE_DS4 311
#define NOTE_E4 330
#define NOTE_F4 349
#define NOTE_FS4 370
#define NOTE_G4 392
#define NOTE_GS4 415
#define NOTE_A4 440
#define NOTE_AS4 466
#define NOTE_B4 494
#define NOTE_C5 523
#define NOTE_CS5 554
#define NOTE_D5 587
#define NOTE_DS5 622
#define NOTE_E5 659
#define NOTE_F5 698
```

```
#define NOTE_FS5 740  
#define NOTE_G5 784  
#define NOTE_GS5 831  
#define NOTE_A5 880  
#define NOTE_AS5 932  
#define NOTE_B5 988  
#define NOTE_C6 1047  
#define NOTE_CS6 1109  
#define NOTE_D6 1175  
#define NOTE_DS6 1245  
#define NOTE_E6 1319  
#define NOTE_F6 1397  
#define NOTE_FS6 1480  
#define NOTE_G6 1568  
#define NOTE_GS6 1661  
#define NOTE_A6 1760  
#define NOTE_AS6 1865  
#define NOTE_B6 1976  
#define NOTE_C7 2093  
#define NOTE_CS7 2217  
#define NOTE_D7 2349  
#define NOTE_DS7 2489  
#define NOTE_E7 2637  
#define NOTE_F7 2794  
#define NOTE_FS7 2960  
#define NOTE_G7 3136  
#define NOTE_GS7 3322  
#define NOTE_A7 3520  
#define NOTE_AS7 3729  
#define NOTE_B7 3951  
#define NOTE_C8 4186  
#define NOTE_CS8 4435
```

```

#define NOTE_D8 4699

#define NOTE_DS8 4978

int melody[] = {

NOTE_CS4, NOTE_E4, NOTE_CS4, NOTE_CS4, NOTE_E4,

NOTE_CS4, NOTE_CS4, NOTE_E4, NOTE_CS4, NOTE_DS4,

NOTE_CS4, NOTE_CS4, NOTE_E4, NOTE_CS4,

NOTE_B3,

NOTE_CS4, NOTE_E4, NOTE_CS4, NOTE_CS4, NOTE_E4,

NOTE_CS4, NOTE_DS4, NOTE_CS4, NOTE_E4,

NOTE_B3,

NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4,

NOTE_E4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_GS4,

NOTE_GS4, NOTE_E4, NOTE_FS4, NOTE_B4, NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_GS4,

NOTE_GS4,

NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_FS4,

NOTE_FS4, NOTE_GS4, NOTE_FS4,

NOTE_E4, NOTE_CS4, NOTE_CS4, NOTE_GS4, NOTE_GS4,

NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_GS4,

NOTE_GS4,

NOTE_GS4, NOTE_B4, NOTE_GS4, NOTE_GS4, NOTE_FS4, NOTE_FS4, NOTE_E4, NOTE_GS4,

NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_B4, NOTE_GS4, NOTE_GS4, NOTE_GS4, NOTE_FS4,

NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_FS4, NOTE_FS4,

NOTE_E4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_CS4,

NOTE_GS3, NOTE_B3,

NOTE_CS4, NOTE_CS4, NOTE_FS4, NOTE_GS4, NOTE_E4, NOTE_FS4,

NOTE_B3,

NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_CS4, NOTE_E4, NOTE_GS4,

NOTE_FS4, NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_B4, NOTE_GS4, NOTE_GS4, NOTE_FS4,

NOTE_FS4,

NOTE_E4, NOTE_CS4, NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4,

NOTE_FS4, NOTE_E4, NOTE_CS4, NOTE_CS4,

NOTE_B3,

NOTE_CS4, NOTE_CS4, NOTE_FS4, NOTE_GS4, NOTE_E4, NOTE_FS4, NOTE_FS4,

```

```

NOTE_B3,
NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_CS4, NOTE_E4, NOTE_GS4, NOTE_FS4,
NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_B4, NOTE_GS4, NOTE_GS4, NOTE_FS4, NOTE_FS4, NOTE_E4,
NOTE_CS4, NOTE_B4, NOTE_GS4, NOTE_GS4, NOTE_FS4, NOTE_FS4, NOTE_E4,
NOTE_CS4, NOTE_CS4,
NOTE_GS3, NOTE_B3,
NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_FS4,
NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_E4, NOTE_FS4|
NOTE_CS4, NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_CS4,
NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_FS4, NOTE_E4, NOTE_FS4, NOTE_GS4,
NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_CS4,
NOTE_FS4, NOTE_GS4, NOTE_B4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_FS4,
NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_FS4,
NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_E4, NOTE_FS4, NOTE_GS4,
NOTE_CS4, NOTE_E4,
NOTE_FS4, NOTE_E4, NOTE_FS4, NOTE_E4, NOTE_FS4, NOTE_GS4,
NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_E4, NOTE_E4,
NOTE_FS4, NOTE_FS4, NOTE_GS4, NOTE_GS4,
NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_CS4,
NOTE_CS4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_FS4, NOTE_GS4,
NOTE_GS4, NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4,
NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_E4, NOTE_E4,
NOTE_FS4, NOTE_FS4, NOTE_GS4, NOTE_GS4,
NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_FS4, NOTE_E4, NOTE_E4, NOTE_FS4, NOTE_CS4, NOTE_CS4,
NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_CS4, NOTE_E4, NOTE_FS4, NOTE_GS4, NOTE_CS4,
NOTE_E4, NOTE_FS4,
NOTE_CS4,
};

int durations[] = {
2,2,4,2,2,4,2,2,4,2,2,2,2,2,2,2,2,2,4,2,2,4,2,2,2,2,8,8,4,4,8,8,8,8,8,8,2,8,8,4,2,8,8,4,8,8,8,4,4,8,8,8,8,8,
4,8,4,8,8,8,8,4,2,8,8,8,8,4,8,8,8,8,8,2,8,4,8,8,8,8,8,2,4,8,8,2,4,8,8,8,8,8,2,8,8,4,4,4,8,8,2,4,4,2,4,2,
2,4,8,8,8,2,2,8,4,8,4,4,4,8,8,8,2,8,8,4,8,8,4,2,8,4,8,4,4,4,4,2,2,2,2,4,8,8,8,2,2,2,4,8,4,4,4,4,8,8,8,8,
2,8,8,4,8,8,4,2,8,8,4,8,8,4,2,2,4,2,8,8,4,8,8,4,8,8,2,8,8,4,8,8,4,8,8,2,8,8,4,4,4,8,8,2,8,8,4,8,8,8,4,8,2,4,

```

```
4,8,8,4,4,8,2,8,8,4,4,4,8,8,4,8,8,8,8,8,8,4,4,4,4,8,2,8,8,4,8,8,8,4,8,4,4,4,4,4,4,2,8,8,4,8,8,8,4,8,4,4,4,
4,4,4,4,2,8,8,4,8,8,8,4,8,4,4,4,4,4,4,2,8,8,4,8,8,8,4,8,4,8,8,8,8,8,4,4,4,4,2
```

```
};
```

```
void setup()
```

```
{
```

```
pinMode(BUZZER_PIN, OUTPUT);
```

```
}
```

```
void loop()
```

```
{
```

```
int size = sizeof(durations) / sizeof(int);
```

```
for (int note = 0; note < size; note++) {
```

```
//to calculate the note duration, take one second divided by the note type.
```

```
//e.g. quarter note = 1000 / 4, eighth note = 1000/8, etc.
```

```
int duration = 1000 / durations[note];
```

```
tone(BUZZER_PIN, melody[note], duration);
```

```
//to distinguish the notes, set a minimum time between them.
```

```
//the note's duration + 30% seems to work well:
```

```
int pauseBetweenNotes = duration * 1.30;
```

```
delay(pauseBetweenNotes);
```

```
//stop the tone playing:
```

```
noTone(BUZZER_PIN);
```

```
}
```

```
}
```

Das ganze funktioniert sehr einfach.